# Complete Technical Documentation

Frontend, Backend Architecture, Database Schema, Machine Learning & Recent Enhancements

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|---|---|
| **Project:** AI Predictive Personnel Stress (APPS) | **Environment:** Vercel Serverless / Cloud Blob |
| **Backend:** Python Flask 3.x / Scikit-Learn | **Database:** SQLite3 + Two-Way Cloud Blob Sync |
| **Frontend:** Responsive HTML5 / CSS3 / Jinja2 | **Status:** Production Live Verified |
| **Live URL:** https://apps-nu-henna.vercel.app | **Generated:** October 2026 \| Version 2.0 |

## 1. Executive Summary & System Overview

**APPS (AI Predictive Personnel Stress - Welfare and Monitoring System)** is a full-stack, mission-critical digital psychological welfare platform designed to monitor, assess, and support the mental wellbeing of military personnel. The system incorporates an automated **Machine Learning Random Forest classifier** to analyze daily physiological and operational metrics (sleep duration, active duty hours, resting heart rate, and mood score), flagging soldiers experiencing heightened stress levels and delivering personalized coping recommendations.

The platform establishes a strict, role-governed operational hierarchy among three distinct user categories: **(1) Personnel (Soldiers)** who submit daily check-ins and request assistance, **(2) Welfare Officers** who manage assigned soldiers, monitor counseling logs, review clinical notes, and handle triage requests, and **(3) Senior Administrators** who monitor force-wide wellness trends, unit-level risk distributions, generate access credentials, and export audit-ready reports.

| Core Module | Primary Responsibility | Technologies & Protocols |
|---|---|---|
| **Frontend Client** | Adaptive UI for smartphones, tablets, and laptops. Role-based dashboards, survey forms, live password strength indicators. | HTML5, Custom Glassmorphic CSS3, Vanilla JS, Jinja2 Templates |
| **Backend Engine** | REST routing, role decorators (RBAC), CSRF & rate limit protections, automated officer load-balancing, ML inference. | Python, Flask, Werkzeug Security, Scikit-learn, Joblib |
| **Predictive AI** | Predicts stress levels (Low, Moderate, High) from physiological telemetry; generates automated non-clinical wellness advice. | Scikit-learn Random Forest Classifier, Pandas, NumPy |
| **Database & Cloud Sync** | Relational persistence with real-time bidirectional synchronization between local SQLite and Vercel Blob Object Storage. | SQLite3, Vercel Blob Private REST API, SyncedConnection |
| **Audit & Reporting** | Generates unit stress analytics, Excel spreadsheets with injection defense, sanitized PDFs, and CSV telemetry exports. | ReportLab, OpenPyXL, CSV, ActivityLog audit trail |

## 2. Frontend Architecture & User Interface Design

The APPS frontend is architected as a lightweight, lightning-fast, zero-dependency server-side rendered application. It avoids bloated JavaScript frameworks in favor of clean semantic HTML5, Jinja2 templates, and a custom CSS design system built with modern CSS variables, responsive flexbox/grid layouts, and glassmorphic aesthetic cards.

## A. Mobile-First & Desktop Ergonomics (Responsive Design)

A major overhaul was deployed to guarantee optimal usability across all form factors: mobile smartphones, tablets, and desktop workstations.

• **Desktop View (>920px):** Expansive horizontal navigation bar, multi-column metric cards, full tabular summaries, and prominent quick-action toolbars.

• **Mobile View (≤920px & ≤480px):** Automated conversion to an off-canvas slide-out hamburger navigation drawer, stacked single-column cards, horizontal swipe containers for data tables, and enlarged 48px touch targets for thumb ergonomics.

• **Fluid Layout Engine:** Responsive typography using viewport clamps, flexible container margins (14px on mobile up to 1200px centered on desktop), and mobile-optimized login shells preventing iOS/Android keyboard zoom.

## B. Complete Template Catalog & View Hierarchy

| Template File | Route / URL | UI Purpose & Key Components |
|---|---|---|
| `portal.html` | `/portal, /` | Unified entry landing page. Features role selection cards (Soldier, Officer, Administrator) with explicit signup links. |
| `login.html` | `/login, /*_login` | Single Unified Login interface with role selector tabs, remember-me support, password visibility toggle, and error toasts. |
| `signup.html` | `/signup` | Personnel registration form. Unit selection, live password strength requirements meter, and officer auto-assignment notice. |
| `officer_signup.html` | `/officer_signup` | Officer registration. Requires cryptographically verified one-time officer access code (OFC-XXXXXX). |
| `admin_signup.html` | `/admin_signup` | Senior Administrator registration. Requires one-time administrator access code (ADM-XXXXXX). |
| `upgrade_password.html` | `/upgrade_password` | Mandatory password upgrade screen for users with legacy/weak passwords. Enforces minimum length and complexity. |
| `dashboard.html` | `/dashboard` | Personnel daily portal. Daily wellness survey input, 7-day health trend chart, recent check-in history, quick stress relief button. |
| `results.html` | `/results` | Immediate post-checkin feedback. Shows AI-predicted stress level badge (Low/Moderate/High) and tailored coping advice. |
| `snapshot.html` | `/snapshot` | Personnel historical health telemetry view with PDF export trigger. |
| `officer_dashboard.html` | `/officer_dashboard` | Welfare Officer primary station. Summary counters, urgent triage alert queue, going quiet counter, workload indicators. |
| `officer_personnel.html` | `/officer/personnel` | Single-card roster view of all assigned soldiers. Displays unit badge, last check-in timestamp, stress tag, and profile link. |
| `personnel_detail.html` | `/officer/personnel/<id>` | Deep-dive soldier dossier. Physiological time-series charts, counseling note authoring modal, risk tags manager. |
| `officer_going_quiet.html` | `/officer/going_quiet` | Early-warning monitoring page listing personnel who have not submitted a check-in for 3 or more consecutive days. |
| `officer_requests.html` | `/officer/requests` | 5-stage stress relief request kanban/pipeline. Officers can advance status from New → Acknowledged → In Progress → Resolved. |
| `officer_activity_log.html` | `/officer/activity_log` | Immutable audit trail displaying officer actions, note additions, tag updates, and status transitions. |
| `admin_dashboard.html` | `/admin_dashboard` | Senior Command Command Center. Unit stress distribution breakdown, high-risk flags, one-time code generator, batch export buttons. |
| `admin_units.html` | `/admin/units` | Comparative unit welfare index table comparing risk percentages across regiments, battalions, and commands. |

# 3. Backend Architecture & API Specifications

The backend is powered by **Python 3 and Flask**, engineered for serverless deployment on Vercel Functions. It employs modular architectural patterns including custom request decorators, strict security policies, cryptographic password hashing, rate limiting, and an automated officer assignment load balancer.

## A. Key Architectural Subsystems

• **Role-Based Access Control (RBAC):** Implemented via @login_required and @role_required('admin'|'officer'|'personnel') decorators. Unauthorized requests are redirected to role-specific login routes with contextual flash error notices.
• **Automated Officer Load Balancer (pick_officer):** When a new soldier signs up, the system executes an atomic subquery calculating the active caseload of all registered officers and assigns the soldier to the officer with the lowest workload, ensuring fair distribution across medical staff.
• **CSRF & Same-Origin Enforcement:** All state-mutating POST requests undergo header origin checks verifying the request origin matches the host domain.
• **Rate Limiting & Brute-Force Backoff:** Failed login attempts are throttled to prevent credential stuffing attacks.
• **Formula & CSV Injection Defense:** When generating Excel and CSV exports, strings prefixed with arithmetic characters (=, +, -, @) are sanitized to prevent remote code execution in spreadsheet software.

## B. Complete Route & API Reference

| Route Path | HTTP | Role | Functionality & Behavior |
|---|---|---|---|
| `/login` | GET/POST | Public | Authenticates users, checks password strength; redirects to upgrade or dashboard. |
| `/signup` | GET/POST | Public | Creates soldier account; enforces strong password; calls pick_officer(). |
| `/officer_signup` | GET/POST | Public | Validates and consumes single-use OFC-* code; creates officer account. |
| `/admin_signup` | GET/POST | Public | Validates and consumes single-use ADM-* code; creates admin account. |
| `/upgrade_password` | GET/POST | Auth | Upgrades legacy/weak password to PBKDF2 hash and syncs to cloud storage. |
| `/submit_survey` | POST | Soldier | Validates inputs; invokes Random Forest ML model; writes DailyData record. |
| `/submit_stress_request` | POST | Soldier | Dispatches confidential stress relief request to assigned officer. |
| `/officer/requests/<id>/advance` | POST | Officer | Transitions request through stage pipeline (New → Acknowledged → Resolved). |
| `/add_note` | POST | Officer | Saves confidential clinical counseling note and records audit ActivityLog. |
| `/add_tag` | POST | Officer | Attaches risk/status classification tag (e.g. #HighStress) to soldier dossier. |
| `/admin/generate_code` | POST | Admin | Generates cryptographically random one-time registration codes for staff. |
| `/admin/export_excel` | GET | Admin | Generates styled OpenPyXL workbook containing force-wide health data. |
| `/officer/export_report` | GET | Officer | Compiles ReportLab PDF wellness dossier for assigned personnel. |

# 4. Machine Learning & Stress Prediction Subsystem

The core diagnostic engine of APPS utilizes an offline-trained **Random Forest Classifier** serialized with Joblib (stress_model.pkl). During every daily survey submission, the model evaluates multidimensional physiological and operational indicators in real-time to deliver an immediate risk classification without human latency.

| Feature Column | Data Type | Valid Range | Clinical Correlation & Significance |
|---|---|---|---|
| `sleep_hours` | Float (REAL) | 0.0 – 24.0 hrs | Restorative rest. Sleep deprivation (<5 hrs) strongly correlates with cognitive fatigue and acute stress. |
| `duty_hours` | Float (REAL) | 0.0 – 24.0 hrs | Operational workload. Extended duty shifts (>14 hrs) increase burn-out potential and cardiovascular strain. |
| `heart_rate` | Integer | 40 – 200 bpm | Resting / baseline pulse. Elevated resting heart rate serves as an autonomic indicator of sympathetic nervous activation. |
| `mood_score` | Integer | 1 (Low) – 10 (High) | Subjective emotional state self-reported by personnel. Strong inverse predictor of depressive/anxious symptoms. |

**Model Inference & Actionable Outputs:**
• **Low Stress:** Positive reinforcement, suggestions for maintaining physical training and good sleep hygiene.
• **Moderate Stress:** Non-clinical intervention advice, including 4-7-8 breathing exercises, tactical hydration, and micro-breaks.
• **High Stress:** Immediate alert badge, priority referral prompts to assigned Welfare Officer, and automated inclusion in officer urgent triage queues.

# 5. Database Architecture & Schema Specification

The application utilizes a normalized relational schema executed on **SQLite3**. The schema isolates sensitive military medical telemetry, officer clinical notes, authentication tokens, and user credentials.

| Table Name | Key Columns & Constraints | Description & Relationship |
|---|---|---|
| `Users` | `id (PK), name, unit, password, role, assigned_officer_id (FK)` | Stores personnel credentials (PBKDF2 hashes), regiment/unit, assigned role, and linked welfare officer. |
| `DailyData` | `id (PK), user_id (FK), date, sleep_hours, duty_hours, heart_rate, mood_score, stress_level, recommendation` | Daily wellness check-ins. One record per user per day. Stores physiological telemetry and AI predictions. |
| `OfficerCodes` | `id (PK), code (UNIQUE), is_used (0\|1)` | Cryptographically seeded one-time registration codes (OFC-XXXXXX) for authorized medical officers. |
| `AdminCodes` | `id (PK), code (UNIQUE), is_used (0\|1)` | Single-use registration tokens (ADM-XXXXXX) for system administrators. |
| `StressRequests` | `id (PK), user_id (FK), message, status, stage, officer_id (FK), remedy, created_date, resolved_date, stage_updated_at` | Confidential assistance requests tracking through a 5-stage lifecycle (New → Acknowledged → In Progress → Resolved → Closed). |
| `CounselingNotes` | `id (PK), user_id (FK), officer_id (FK), date, notes, action_taken, follow_up_date` | Privileged clinical case notes maintained by welfare officers, including follow-up dates. |
| `ActivityLog` | `id (PK), actor_id (FK), action, target_user_id (FK), details, created_at` | Immutable audit trail logging all officer and admin actions for accountability and forensic integrity. |
| `PersonnelTags` | `id (PK), user_id (FK), tag (NOCASE), officer_id (FK), created_date, UNIQUE(user_id, tag)` | Custom tags applied by officers (e.g. #HighRisk, #FollowUpRequired) for rapid categorization. |

## 6. Serverless Cloud Persistence & Vercel Blob Sync Engine

One of the most critical engineering challenges in serverless architectures is **filesystem ephemerality**. On Vercel, serverless function containers are destroyed and recreated frequently (container cold boots). Any SQLite database stored on /tmp/apps.db is wiped back to the git bundle on container restart, causing newly updated passwords, signups, and check-ins to vanish after idle periods.

To permanently rectify this, a custom **Bidirectional Cloud Object Storage Synchronization Engine** was engineered.

**ARCHITECTURAL INNOVATION: SyncedConnection & Cloud Hydration**

1. Automatic Cold-Boot Hydration: On container startup, sync_db_from_blob() checks the private Vercel Blob store ('apps-data'). If a newer master apps.db exists, it is securely downloaded and hydrated into the container. 2. Transparent Commit Hook: By subclassing sqlite3.Connection into SyncedConnection and injecting it as the connection factory, every commit() automatically triggers sync_db_to_blob(), pushing instantaneous database snapshots to private blob storage. 3. Zero-Data-Loss Guarantee: Updates to passwords, new users, and counseling notes survive serverless container recycles indefinitely.

## 7. Key Enhancements & Changelog (Recent Additions)

The following itemized features and structural enhancements were designed, coded, and deployed to production:

| Category | Enhancement / Feature Added | Impact & Operational Benefit |
|---|---|---|
| **Security & Auth** | **Weak Password Policy & Forced Upgrade Flow:** Enforced minimum 6-character length and complexity rules. Banned trivial passwords (123, 123456, password). Built mandatory /upgrade_password route that intercepts existing users with legacy passwords upon login. | Completely eliminated vulnerabilities from default weak credentials; upgraded existing user accounts securely to PBKDF2 hashes. |
| **Data Persistence** | **Vercel Blob Storage Integration:** Provisioned private store apps-data, configured access tokens, built SyncedConnection and automatic two-way synchronization in app.py. | Resolved the serverless container reset bug where password updates disappeared after overnight cold restarts. |
| **Responsive Design** | **Full Mobile & Laptop Responsive Layout:** Engineered adaptive breakpoints (860px, 720px, 480px). Added mobile slide drawer navigation, stacked stat cards, responsive tables with horizontal scrolling, and touch-friendly 48px buttons. | Seamless user experience across smartphones, iPads/tablets, and high-resolution laptop monitors. |
| **User Experience** | **Single Portal & Dedicated Signup Pathways:** Streamlined entry page (/portal & /login) with clear role cards and explicit, visible signup links for new Officers and Admins. | Eliminated user confusion regarding how new officers and administrators can register with their authorized access codes. |
| **Officer Dashboard** | **Personnel Roster Single-Card Redesign:** Redesigned the soldier list in officer_personnel.html so each person occupies a clean, single card with inline unit badges and last check-in timestamp. | Eliminated visual clutter and disorganized stacked text, presenting medical staff with a clear clinical roster. |
| **Clinical Safety** | **'Going Quiet' Early Warning Watchlist:** Automated identification of soldiers who have missed daily check-ins for 3+ consecutive days, surfaced on a dedicated priority monitoring page. | Enables welfare officers to proactively reach out to at-risk soldiers before mental distress escalates unnoticed. |
| **Workflow Tracking** | **Universal 5-Stage Stress Request Pipeline:** Transformed requests into an omnipresent triage pipeline accessible to all on-duty officers. Requests from all personnel dispatch directly to the Officer Dashboard with real-time pending badges and alert banners. | Ensures no personnel message is missed, enabling collaborative review and transparent status tracking. |
| **Reporting & Exports** | **Multi-Format Data Exports (PDF / Excel / CSV):** Integrated ReportLab PDF generation, OpenPyXL spreadsheet generation with formula injection protection, and telemetry CSV downloads. | Enables senior officers and medical boards to produce official physical dossiers and command briefing sheets. |

# 8. Deployment & Production Verification

| Parameter | Production Configuration Value |
|---|---|
| **Production Web URL** | `https://apps-nu-henna.vercel.app` |
| **Hosting Platform** | Vercel Serverless Platform (Node.js/Python WSGI Bridge) |
| **Object Storage Store** | Vercel Blob Storage • Store: apps-data (ID: store_eS9r7map43Dx32yX) |
| **Source Control** | Git (Dual-tracked: origin/main and origin/master) |
| **Secret Key Management** | Environment-backed session secret key with stable production fallback |
| **Database Verification** | Verified live: User 'Dharun' upgraded to strong password, persistent across cold boots |

**DOCUMENT CONCLUSION & SYSTEM HEALTH STATUS**

The APPS platform is fully operational, verified in production, and hardened against security vulnerabilities, data loss on cold boots, and layout distortions across mobile and desktop devices. All features detailed in this specification are live and accessible.